

Abstract No.

Deciphering Ceramide Biosynthesis as a Metabolic Dependency in MYC-Driven Group 3 Medulloblastoma

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Medulloblastoma (MB) is the most common malignant paediatric brain tumour, classified into four molecular subgroups with distinct biology and prognosis. Among these, MYC-amplified Group 3 MB (MB Grp3) represents one of the highest-risk variants, with 5-year survival below 40% despite aggressive multimodal treatment that leads to severe long-term neurocognitive and endocrine sequelae.

Metabolic reprogramming has emerged as a key cancer hallmark driving tumour progression and treatment resistance, making subgroup-specific metabolic vulnerabilities a promising, largely unexplored therapeutic strategy in high-risk MB. To this end, we profiled mouse models of Myc-driven and Shh TP53-mutant MB by bulk RNA-sequencing and untargeted metabolomics. Myc-amplified MB Grp3 displayed higher lipid accumulation, predominantly sphingolipids (SL), than Shh p53mut MBs and controls.

Metabolic enzymes were characterized across complementary human models, including patient-derived orthotopic xenografts (PDOXs), FFPE specimens, PDOX-derived MB stem cells (MBSCs) and their xenografts, and MB human cell lines. Pharmacological MYC inhibition (10058-F4) and radioactive metabolic flux assays were used to probe pathway regulation and activity.

The ceramide (CER) biosynthetic metabolism emerged as a distinguishing feature, with Myc MB-restricted upregulation of enzymes like SPTLC1, CERS2, and CERS4. CERS4 was the most robust, subgroup-specific isoform, and its expression was selectively reduced upon pharmacological MYC inhibition in human MYC MB Grp3 cells, but not in SHH cells, supporting direct MYC-dependent regulation. Metabolic flux assays in human MBSCs confirmed enhanced CER synthesis and conversion into complex CER species in MYC- versus SHH-driven cells. As CER can act as either pro-apoptotic or pro-survival signal depending on its downstream fate, we confirmed that pharmacologically shifting this balance selectively impairs MYC MBSC viability, supporting the functional relevance of the pathway.

These data identify CER biosynthesis, and CERS4 in particular, as a MYC-regulated, subgroup-selective metabolic dependency in high-risk MB Grp3, revealing novel druggable vulnerabilities for precision therapy in this aggressive paediatric tumor.



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